

Robust Myelination of Regenerated Axons Induced by Combined Manipulations of GPR17 and Microglia.

PMID: 33108748 | Indexed in PubMed

Myelination facilitates rapid axonal conduction, enabling efficient communication across different parts of the nervous system. Here we examined mechanisms controlling myelination after injury and during axon regeneration in the central nervous system (CNS). Previously, we discovered multiple molecular pathways and strategies that could promote robust axon regrowth after optic nerve injury. However, regenerated axons remain unmyelinated, and the underlying mechanisms are elusive. In this study, we found that, in injured optic nerves, oligodendrocyte precursor cells (OPCs) undergo transient proliferation but fail to differentiate into mature myelination-competent oligodendrocytes, reminiscent of what is observed in human progressive multiple sclerosis. Mechanistically, we showed that OPC-intrinsic GPR17 signaling and sustained activation of microglia inhibit different stages of OPC differentiation. Importantly, co-manipulation of GPR17 and microglia led to extensive myelination of regenerated axons. The regulatory mechanisms of stage-dependent OPC differentiation uncovered here suggest a translatable strategy for efficient de novo myelination after CNS injury.

Axo-myelinic neurotransmission: a novel mode of cell signalling in the central nervous system.

PMID: 29118449 | Indexed in PubMed

It is widely recognized that myelination of axons greatly enhances the speed of signal transmission. An exciting new finding is the dynamic communication between axons and their myelin-forming oligodendrocytes, including activity-dependent signalling from axon to myelin. The oligodendrocyte-myelin complex may in turn respond by providing metabolic support or alter subtle myelin properties to modulate action potential propagation. In this Opinion, we discuss what is known regarding the molecular physiology of this novel, synapse-like communication and speculate on potential roles in disease states including multiple sclerosis, schizophrenia and Alzheimer disease. An emerging appreciation of the contribution of white-matter perturbations to neurological dysfunction identifies the axo-myelinic synapse as a potential novel therapeutic target.

Induction of myelination in the central nervous system by electrical activity.

PMID: 8790426 | Indexed in PubMed

The oligodendrocyte is the myelin-forming cell in the central nervous system. Despite the close interaction between axons and oligodendrocytes, there is little evidence that neurons influence myelinogenesis. On the contrary, newly differentiated oligodendrocytes, which mature in culture in the total absence of neurons, synthesize the myelin-specific constituents of oligodendrocytes differentiated in vivo and even form myelin-like figures. Neuronal electrical activity may be required, however, for the appropriate formation of the myelin sheath. To investigate the role of electrical activity on myelin formation, we have used highly specific neurotoxins, which can either block (tetrodotoxin) or increase (alpha-scorpion toxin) the firing of neurons. We show that myelination can be inhibited by blocking the action potential of neighboring axons or enhanced by increasing their electrical activity, clearly linking

neuronal electrical activity to myelinogenesis.

Glutamate versus GABA in neuron-oligodendroglia communication.

PMID: 30957306 | Indexed in PubMed

In the central nervous system (CNS), myelin sheaths around axons are formed by glial cells named oligodendrocytes (OLs). In turn, OLs are generated by oligodendrocyte precursor cells (OPCs) during postnatal development and in adults, according to a process that depends on the proliferation and differentiation of these progenitors. The maturation of OL lineage cells as well as myelination by OLs are complex and highly regulated processes in the CNS. OPCs and OLs express an array of receptors for neurotransmitters, in particular for the two main CNS neurotransmitters glutamate and GABA, and are therefore endowed with the capacity to respond to neuronal activity. Initial studies in cell cultures demonstrated that both glutamate and GABA signaling mechanisms play important roles in OL lineage cell development and function. However, much remains to be learned about the communication of glutamatergic and GABAergic neurons with oligodendroglia in vivo. This review focuses on recent major advances in our understanding of the neuron-oligodendroglia communication mediated by glutamate and GABA in the CNS, and highlights the present controversies in the field. We discuss the expression, activation modes and potential roles of synaptic and extrasynaptic receptors along OL lineage progression. We review the properties of OPC synaptic connectivity with presynaptic glutamatergic and GABAergic neurons in the brain and consider the implication of glutamate and GABA signaling in activity-driven adaptive myelination.

Chromatin remodelers in oligodendroglia.

PMID: 32460418 | Indexed in PubMed

Oligodendrocytes, the myelinating cells in the vertebrate central nervous system, produce myelin sheaths to enable saltatory propagation of action potentials. The process of oligodendrocyte myelination entails a stepwise progression from precursor specification to differentiation, which is coordinated by a series of transcriptional and chromatin remodeling events. ATP-dependent chromatin remodeling enzymes, which utilize ATP as an energy source to control chromatin dynamics and regulate the accessibility of chromatin to transcriptional regulators, are critical for oligodendrocyte lineage development and regeneration. In this review, we focus on the latest insights into the spatial and temporal specificity of chromatin remodelers during oligodendrocyte development, myelinogenesis, and regeneration. We will also bring together various plausible mechanisms by which lineage specific transcriptional regulators coordinate with chromatin remodeling factors for programming genomic landscapes to specifically modulate these different processes during developmental myelination and remyelination upon injury.

Pathology of oligodendroglia: An overview.

PMID: 28548216 | Indexed in PubMed

Oligodendroglia are cells responsible for creating myelin sheaths for axons in the CNS. However, pathologies of oligodendroglia other than demyelination are not well understood due to the lack of adequate methods of characterizing pathological conditions affecting oligodendroglia in human tissue. This review discusses three major topics with the aim of clarifying some of the controversies in the study of oligodendroglia. The oligodendroglioma, a relatively indolent form of diffuse gliomas thought to originate in oligodendrocytes, has never demonstrated myelin formation on electron microscopy nor shown a constant expression of myelin-related proteins. Oligodendrogliomas instead share an immune phenotype with oligodendrocyte progenitor cells (OPCs). Another type of cell that resembles OPCs are oligodendroglia-like cells (OLCs), which occur in many types of low-grade tumors and focal cortical dysplasia. In neurodegenerative disorders, oligodendroglia can be a target of abnormal aggregations of proteins such as tau. Tau-positive oligodendroglial inclusions in progressive supranuclear palsy and corticobasal generation differ from each other morphologically, ultrastructurally and biochemically, suggesting disparate underlying pathological processes despite significant overlapping of the clinical manifestations. To promote the study of oligodendroglia, novel methods for detecting OLCs in situ are urgently required.

Oligodendrocytes in central nervous system diseases: the effect of cytokine regulation.

PMID: 38488548 | Indexed in PubMed

Cytokines including tumor necrosis factor, interleukins, interferons, and chemokines are abundantly produced in various diseases. As pleiotropic factors, cytokines are involved in nearly every aspect of cellular functions such as migration, survival, proliferation, and differentiation. Oligodendrocytes are the myelin-forming cells in the central nervous system and play critical roles in the conduction of action potentials, supply of metabolic components for axons, and other functions. Emerging evidence suggests that both oligodendrocytes and oligodendrocyte precursor cells are vulnerable to cytokines released under pathological conditions. This review mainly summarizes the effects of cytokines on oligodendrocyte lineage cells in central nervous system diseases. A comprehensive understanding of the effects of cytokines on oligodendrocyte lineage cells contributes to our understanding of central nervous system diseases and offers insights into treatment strategies.

Crazy Little Thing Called Sox-New Insights in Oligodendroglial Sox Protein Function.

PMID: 31159496 | Indexed in PubMed

In the central nervous system, oligodendrocytes wrap axons with myelin sheaths, which is essential for rapid transfer of electric signals and their trophic support. In oligodendroglia, transcription factors of the Sox protein family are pivotal regulators of a variety of developmental processes. These include specification, proliferation, and migration of oligodendrocyte precursor cells as well as terminal differentiation to mature myelinating oligodendrocytes. Sox proteins are further affected in demyelinating diseases and are involved in remyelination following damage of the central nervous system. Here we summarize and discuss latest findings on transcriptional regulation of Sox proteins, their function, target genes, and interaction with other transcription factors and chromatin remodelers in oligodendroglia with physiological and pathophysiological relevance.

Hematopoietic progenitors express myelin basic protein and ensheath axons in Shiverer brain.

PMID: 23375677 | Indexed in PubMed

Oligodendroglia are cells of the central nervous system (CNS) that form myelin sheath, which insulates neuronal axons. Neuropathologies of the CNS include dysmyelination of axons in multiple sclerosis and CNS trauma. Cell replacement is a promising but largely untested therapy for dysmyelination. Shiverer mouse, a genetic mutant that does not synthesize full-length myelin basic protein (MBP), a critical prerequisite protein in CNS myelin sheath formation, provides an unequivocal model for determining the potential of stem cells to become oligodendroglia. We demonstrate that adult wild-type mouse bone marrow stem cells can express MBP and ensheath axons when transplanted into Shiverer brain.

The polarity protein Scribble regulates myelination and remyelination in the central nervous system.

PMID: 25807062 | Indexed in PubMed

The development and regeneration of myelin by oligodendrocytes, the myelin-forming cells of the central nervous system (CNS), requires profound changes in cell shape that lead to myelin sheath initiation and formation. Here, we demonstrate a requirement for the basal polarity complex protein Scribble in CNS myelination and remyelination. Scribble is expressed throughout oligodendroglial development and is up-regulated in mature oligodendrocytes where it is localised to both developing and mature CNS myelin sheaths. Knockdown of Scribble expression in cultured oligodendroglia results in disrupted morphology and myelination initiation. When Scribble expression is conditionally eliminated in the myelinating glia of transgenic mice, myelin initiation in CNS is disrupted, both during development and following focal demyelination, and longitudinal extension of the myelin sheath is disrupted. At later stages of myelination, Scribble acts to negatively regulate myelin thickness whilst suppressing the extracellular signal-related kinase (ERK)/mitogen-activated protein kinase (MAP) kinase pathway, and localises to non-compact myelin flanking the node of Ranvier where it is required for paranodal axo-glial adhesion. These findings demonstrate an essential role for the evolutionarily-conserved regulators of intracellular polarity in myelination and remyelination.